Regular Analysis Through Continuous Exercise

Ellipse : RACE-01

Single Correct Type Questions:

1. Find the equation to the ellipse, whose focus is the point (−1,1), whose directrix is the straight line $x - y + 3 = 0$, and whose eccentricity is $\frac{1}{2}$.

(A) $5x^2 + 2xy + 7y^2 + 10x - 10y + 7 = 0$
(B) $7x^2 + 2xy + 5y^2 + 10x - 10y + 7 = 0$
(C) $7x^2 + 2xy + 7y^2 + 10x - 10y + 7 = 0$
(D) $7x^2 + 2xy + 7y^2 + 8x - 10y + 7 = 0$

MEL135

2. If LR of an ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, $(a < b)$ is half of its major axis, its eccentricity is.

(A) $e=\frac{1}{\sqrt{3}}$ (B) $e=\frac{1}{\sqrt{2}}$ (C) $e=\frac{1}{\sqrt{5}}$ (D) $e=\frac{1}{\sqrt{7}}$

MEL136

3. The equation of the ellipse whose foci are (4, 6) & (16, 6) and whose semi-minor axis is 4 is-

(A) $\frac{(x-10)^2}{52}+\frac{(y-6)^2}{16}=1$ (B) $\frac{(x-8)^2}{52}+\frac{(y-6)^2}{16}=1$

(C) $\frac{(x-10)^2}{52}+\frac{(y-6)^2}{4}=1$ (D) $\frac{(x-10)^2}{52}+\frac{(y-8)^2}{16}=1$

MEL137

4. Select incorrect statement for ellipse $x^2 + 4y^2 + 8y - 2x + 1 = 0$.

(A) $e=\frac{\sqrt{3}}{2}$ (B) foci $=(1\pm\sqrt{3}, -1)$

(C) $LR=1$ (D) $LR=2$

MEL138

5. The equation to the ellipse, whose centres are the origin, whose axes are the axes of coordinates, and which pass through the points (2,2), and (3,1) is-

(A) $2x^2 + 3y^2 = 20$ (B) $3x^2 + 5y^2 = 32$
(C) $3x^2 + 4y^2 = 31$ (D) $2x^2 + 3y^2 = 21$

MEL139

6. The foci of an ellipse are $(0, \pm 2)$ and its eccentricity is $\frac{1}{\sqrt{2}}$. then its equation is

(A) $\frac{x^2}{2}+\frac{y^2}{8}=1$ (B) $\frac{x^2}{4}+\frac{y^2}{6}=1$ (C) $\frac{x^2}{4}+\frac{y^2}{3}=1$ (D) $\frac{x^2}{4}+\frac{y^2}{8}=1$

MEL140

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7. Select incorrect statement for ellipse $12x^2 + 4y^2 + 24x - 16y + 25 = 0$.

(A) Center $\equiv (-2, 2)$

(B) length of major axis $= \sqrt{3}$, length of minor axis $= 1$

(C) $e = \sqrt{\frac{2}{3}}$

(D) $f\left(-1,\ 2 \pm \frac{1}{\sqrt{2}}\right)$

MEL141

8. The equation $\frac{x^2}{8-t} + \frac{y^2}{t-4} = 1$, will represent an ellipse if

(A) $t \in (1, 5)$ (B) $t \in (2, 8)$ (C) $t \in (4, 8) - \{6\}$ (D) $t \in (4, 10) - \{6\}$

MEL142

9. Which of the following statement is incorrect for the ellipse $2x^2 + 9y^2 = 113$.

(A) Point (4,3) lies on the ellipse (B) Point (4,3) lies outside the ellipse

(C) Point (5,6) lies outside the ellipse (D) Point (1,2) lies inside the ellipse

MEL143

Subjective Type Questions:

10. Find the equation of the ellipse with centre origin and axes as the axes of coordinates

(a) whose latus rectum is 5 and whose eccentricity is $\frac{2}{3}$,

MEL144

(b) whose minor axis is equal to the distance between the foci and whose latus rectum is 10,

MEL145

(c) Whose foci are the points (4,0) and (−4,0) and whose eccentricity is $\frac{1}{3}$

MEL146

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9
Answer	C	B	A	D	B	D	A	C	B

10. (a) $20x^2 + 36y^2 = 405$ (b) $x^2 + 2y^2 = 100$ (c) $8x^2 + 9y^2 = 1152$

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RACE
Regular Analysis Through Continuous Exercise

Ellipse : RACE-02

Single Correct Type Questions:

1. Position of straight line $y = x + \sqrt{\frac{7}{12}}$ with respect to the ellipse $3x^2 + 4y^2 = 1$ is.

(A) Tangent of the ellipse
(B) Secant of ellipse
(C) Normal
(D) Focal Chord

MEL147

2. Line $2tx + y\sqrt{1-t^2} = 1$ touches a fixed ellipse for $\forall t \in \mathbb{R}$ then the eccentricity of the ellipse is-

(A) $\frac{1}{2}$
(B) $\frac{\sqrt{3}}{2}$
(C) $\frac{1}{\sqrt{2}}$
(D) $\frac{1}{\sqrt{3}}$

MEL148

3. Find the condition for the line $x\cos\theta + y\sin\theta = P$ to be a tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

(A) $P^2 = b^2\cos^2\theta + a^2\sin^2\theta$
(B) $P^2 = a^2\cos^2\theta - b^2\sin^2\theta$
(C) $P^2 = a^2\cos^2\theta + b^2\sin^2\theta$
(D) $P^2 = b^2\cos^2\theta - a^2\sin^2\theta$

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MEL149

4. The equation of the tangents to the ellipse $9x^2 + 16y^2 = 144$ which are parallel to the line $x + 3y + k = 0$ is-

(A) $3y + x \pm \sqrt{97} = 0$
(B) $3y + x \pm \sqrt{87} = 0$
(C) $3y + x \pm \sqrt{27} = 0$
(D) $3y + x \pm \sqrt{37} = 0$

MEL150

5. The equation of the tangent to the ellipse $7x^2 + 8y^2 = 100$ at the point (2, –3) is-

(A) $7x - 8y = 50$
(B) $7x + 12y = 50$
(C) $5x - 12y = 50$
(D) $7x - 12y = 50$

MEL151

6. A tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, $(a > b)$ having slope –1 intersects the axis of x & y in point A & B respectively. If O is the origin then find the area of triangle OAB.

(A) $\frac{1}{2}(a^2 + b^2)$
(B) $(a^2 + b^2)$
(C) $\frac{1}{3}(a^2 + b^2)$
(D) $\frac{1}{4}(a^2 + b^2)$

MEL152

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7. Find the lengths of and the equations to, the focal radii drawn to the point $\left(4\sqrt{3},5\right)$ of the ellipse $25x^2 + 16y^2 = 1600$.

(A) $x+4\sqrt{3}y=24\sqrt{3};11x-4\sqrt{3}y=24\sqrt{3}$; 7and 13

(B) $x+4\sqrt{3}y=24\sqrt{3};11x-4\sqrt{3}y=24\sqrt{3}$; 6and 14

(C) $x-4\sqrt{3}y=24\sqrt{3};11x+4\sqrt{3}y=24\sqrt{3}$; 7and 13

(D) $x-4\sqrt{3}y=24\sqrt{3};11x+4\sqrt{3}y=24\sqrt{3}$; 6and 14

MEL153

8. The equation of the normal to the ellipse $9x^2 + 16y^2 = 288$ at the point (4, 3) is-

(A) $4x - 3y = 7$ (B) $2x - 3y = 7$ (C) $4x - 5y = 7$ (D) $3x - 3y = 7$

MEL154

9. If the normal at the point $P(\theta)$ to the ellipse $\frac{x^2}{3}+\frac{y^2}{2}=1$ intersects it again at the point $Q(2\theta)$, then find $\cos\theta$.

(A) –1 (B) 1 (C) 1/2 (D) $\sqrt{\frac{3}{2}}$

MEL155

Subjective Type Questions:

10. Find the equation to the tangent

(a) at the point $\left(1,\frac{4}{3}\right)$ of the ellipse $4x^2 + 9y^2 = 20$.

MEL156

(b) at the point of the ellipse $5x^2 + 3y^2 = 137$ whose ordinate is 2

MEL157

(c) at the ends points of the latus rectum of the ellipse $9x^2 + 16y^2 = 144$.

MEL158

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9
Answer	A	B	C	A	D	A	A	A	A

10. (a) $x + 3y = 5$; (b) $\pm 25x + 6y = 137$; (c) $\pm x\sqrt{7}\pm 4y=16$

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Ellipse : RACE-03

Single Correct Type Questions:

1. If a line $3x - y = 2$ intersects the ellipse $\frac{x^2}{8}+\frac{y^2}{4}=1$ at points A & B, then find co-ordinates of point of intersection of tangents at points A & B.

(A) (12, 2) (B) (–12, –2) (C) (12, –2) (D) (–12, 2)

MEL159

2. Tangents are drawn from the point (–2, 2) to the ellipse $x^2 + 2y^2 = 6$. Find the equation of chord of contact.

(A) $x - 2y + 3 = 0$ (B) $x + 2y + 3 = 0$
(C) $x - 2y - 3 = 0$ (D) $x - 3y + 3 = 0$

MEL160

3. Find the equation of chord of the ellipse $\frac{x^2}{16}+\frac{y^2}{9}=1$ whose mid point be (–1, 1).

(A) $-9x + 16y = 25$ (B) $9x + 16y = 25$
(C) $9x + 16y = -25$ (D) $-9x + 16y = 12$

MEL161

4. The angle between the pair of tangents drawn to the ellipse $2x^2 + 3y^2 = 6$ from the point (–1, 2), is

(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{4}$ (D) $\frac{\pi}{6}$

MEL162

5. Find the equation of chord of contact to the ellipse $\frac{x^2}{16}+\frac{y^2}{9}=1$ of the point (1, 3).

(A) $\frac{x}{16}-\frac{y}{3}=1$ (B) $\frac{y}{16}+\frac{x}{3}=1$ (C) $\frac{x}{14}+\frac{y}{3}=1$ (D) $\frac{x}{16}+\frac{y}{3}=1$

MEL163

6. If chord of contact of the tangent drawn from the point (α, β) to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ touches the circle $x^2 + y^2 = k^2$, then find the locus of the point (α, β).

(A) $\frac{\alpha^2}{a^4}-\frac{\beta^2}{b^2}=\frac{1}{k^2}$ (B) $\frac{\alpha^2}{a^4}+\frac{\beta^2}{b^2}=\frac{1}{k^2}$ (C) $\frac{\alpha^2}{a^2}+\frac{\beta^2}{b^2}=\frac{1}{k^2}$ (D) $\frac{\alpha^2}{a^4}+\frac{\beta^2}{b^4}=\frac{1}{k^2}$

MEL164

7. If the chord of contact of tangents from two points (x_1, y_1) and (x_2, y_2) to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ are at right angles, then find $\frac{x_1x_2}{y_1y_2}$.

(A) $-\frac{a^4}{b^4}$ (B) $\frac{a^4}{b^4}$ (C) $-\frac{a^2}{b^2}$ (D) $\frac{a^2}{b^2}$

MEL165

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8. The locus of the middle points of chords of an ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ which are drawn through the positive end of the minor axis is-

(A) $\frac{x^2}{a^2}+\frac{y^2}{b^2}=\frac{x}{b}$ (B) $\frac{x^2}{a^2}+\frac{y^2}{b^2}=\frac{a}{b}$ (C) $\frac{x^2}{a^2}+\frac{y^2}{b^2}=\frac{y}{b}$ (D) $\frac{x^2}{a^2}+\frac{y^2}{b^2}=\frac{b}{y}$

MEL166

9. The ratio of the area of triangle inscribed in an ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1(a>b)$ to that of triangle formed by the corresponding points on its auxiliary circle is 1/3 then, find the eccentricity of the ellipse.

(A) $\frac{\sqrt{2}}{3}$ (B) $\frac{2\sqrt{2}}{4}$ (C) $\frac{2\sqrt{2}}{5}$ (D) $\frac{2\sqrt{2}}{3}$

MEL167

Integer Type Questions:

10. If F_1 and F_2 are the feet of the perpendiculars from the foci S_1 and S_2, respectively, of an ellipse $\frac{x^2}{25}+\frac{y^2}{16}=1$ on the tangent at any point P on the ellipse, then the minimum value of $S_1F_1 + S_2F_2$ is.

MEL168

11. Line $3x + 4y = 12$ intersect the ellipse $\frac{x^2}{25}+\frac{y^2}{16}=1$ at P and Q. If the point of intersection of tangents at P and Q is $\left(\frac{25}{a},\frac{16}{b}\right)$, then $a + b$ is equal to

MEL169

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Subjective Type Questions:

12. Find the locus of the middle points of chords of an ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$.

(a) whose distance from the centre is the constant length c.

MEL170

(b) which subtend a right angle at the centre.

MEL171

(c) which pass through the given point (h, k)

MEL172

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11
Answer	C	A	A	A	D	D	A	C	D	8	7

12. (a) $(b^2x^2 + a^2y^2)^2 = c^2(b^4x^2 + a^4y^2)$.
(b) $(a^2 + b^2)(b^2x^2 + a^2y^2)^2 = a^2b^2(b^4x^2 + a^4y^2)$.
(c) $b^2x(x - h) + a^2y(y - k) = 0$.

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